

Rohit Lahori

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EDUCATION

Master of Science, Computer Science (AI Specialization)

Illinois Institute of Technology, Chicago, IL

GPA 3.90

May 2026

Coursework: Machine Learning, Natural Language Processing, Big Data Technologies, Deep Learning, Computer Vision

Bachelor of Technology, Computer Science Engineering

Netaji Subhas University of Technology, Delhi, India

GPA 3.14

May 2024

PROGRAMMING SKILLS

- **Languages:** C++, Python, R, SQL, HTML, CSS, Flutter and Dart.
- **Frameworks:** AWS, GCP, MongoDB, FastAPI, Redis, Docker, Qdrant, PostgreSQL, ChromaDb, Kafka, Apache Spark, Power BI
- **ML Library:** TensorFlow, Keras, NLTK, PyTorch, OpenCV, Scikit-Learn, Langchain, LangGraph, MCP, LlamaIndex, Crew AI

WORK EXPERIENCE

AI Developer, Kaplan Institute, Chicago, IL

January 2026 – May 2026

- Engineered a multi-stage student-discovery search engine combining, pgvector semantic search and trigram fuzzy matching fused via Reciprocal Rank Fusion, re-ranked with a Cross-Encoder achieving 45% higher match precision over legacy keyword systems.
- Built a production AI interview platform on AWS with a dedicated LiveKit realtime worker; orchestrated Bedrock and Amazon Nova Sonic for real time voice interviews with context-aware question generation and automated transcript generation via S3.
- Demoed and onboarded 25+ enterprise employers including Grainger and Boeing and 800+ students end-to-end, driving adoption from pilot to production.

AI Engineer Intern, PM Accelerator, Chicago, IL

June 2025 – November 2025

- Launched an AI-powered stock sentiment monitoring platform with a 6-stage agentic pipeline that ingests financial news, analyzes sentiment, and generates real-time trading alerts, slashing false positive alerts by 35%.
- Integrated an agentic retrieval-augmented-generation (RAG) engine, vector-indexing historical sentiment data and auto-validating model outputs, trimming hallucinated responses by $\approx 38\%$ and boosting alert accuracy by 42%.

Data Science Intern, Clootrack Software Labs Pvt Limited, Kochi, India

June 2023 – August 2023

- Improved efficiency of clustering 5,000 interview-based Q&A text data, devising topic labels using various machine learning techniques.
- Enhanced model accuracy to 90% by employing a variety of graph-based (python Louvain and networkx) and hierarchical cluster techniques compared to earlier models in the organization.

RESEARCH PUBLICATIONS

Data Analysis and Insight Generation with Queryable Knowledge Graph (Paper Published in Springer ICICBDA 2024, CCIS 2235)

September 2023 - May 2024

- Implemented a model that leverages data from X (formerly Twitter) and analyses data by segregating tweets into appreciation, complaints, and suggestions through sentiment analysis with 89% precision.
- Transformed raw data into a knowledge graph, uncovering in-depth product intelligence that empowers corporates to make strategic decisions. Integrated Langchain to create recommendations based on insights generated by produced knowledge graph.

TECHNICAL PROJECTS

AI Orchestration Platform - Guidera

May 2025 – August 2025

- Architected an intelligent routing framework for over 50 LLMs, powered by a custom PyTorch-based prompt classifier with a 3-layer attention transformer. This system automated task categorization and deployed a cost-performance optimization algorithm that reduced inference costs by 70% while maintaining 95% accuracy.
- Built a full-scale multimodal AI pipeline that integrates CLIP, OCR, and DenseCap for advanced vision processing. The pipeline includes a two-stage compliance engine that uses GPT-4.1 for policy validation, automatically detects and redacts PII, and analyzes multimodal content to ensure strict adherence to GDPR and HIPAA regulations.

Capsule Hub – Cross Platform AI Context Manager

August 2024 - September 2024

- Shipped a Chrome extension that captures and injects AI conversation context across platforms like ChatGPT, Claude, Gemini Gmail etc, using platform-specific injectors and drag-and-drop, enabling instant context reuse and reducing manual copy-paste.
- Designed backend storage and APIs using MongoDB, FastAPI so teams can save, version, search, and share conversation context with secure login (JWT/OAuth2) and team-based access control.

Real-time plant disease detection on AWS

June 2024 - July 2024

- Created a CNN model to classify plant diseases from images using TensorFlow and Keras with over 38 distinct disease categories. Preprocessed image datasets with data augmentation and executed Conv2D and MaxPooling layers with 94% accuracy.
- Deployed the model on AWS EC2 instance (Ubuntu) and leveraged Streamlit to provide a user-friendly interface for real-time classification results, enhancing both accessibility and predictive visualization.